Parity of an infinitely nested series reversion

A proof of a conjecture in OEIS A195194

Dev D. Goyal

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Abstract

The sequence A195194 of the On-Line Encyclopedia of Integer Sequences (OEIS), contributed by Paul D. Hanna, is defined by an infinitely nested expression,

$$A(x) = \mathrm{INV}\Big(x - x\,\mathrm{INV}\big(x - 2x\,\mathrm{INV}(x - 3x\,\mathrm{INV}(x - \cdots))\big)\Big),$$

where INV denotes series reversion, the operation that undoes a substitution of power series. Its terms are $1, 1, 4, 29, 312, 4528, \ldots$. In 2026 Hanna conjectured that $a(n)$ is odd exactly when n is a power of 2. We first give the infinite nesting a precise meaning: it defines a unique sequence of power series, one for each level, and in particular a unique series A; cutting it off after d levels gives the correct coefficients up to degree $d+1$. This holds for arbitrary weights in place of $1, 2, 3, \ldots$. We then prove the conjecture. Modulo 2 the weight 2 at the second level vanishes, the whole nest collapses, and A satisfies the Catalan equation $A = x + A^2$, whose solution modulo 2 is $x + x^2 + x^4 + x^8 + \cdots$. The results have been checked by computer and proved formally in the Lean proof assistant.

How this paper was produced. This paper reports the result of an AI research effort directed by Dev D. Goyal. ChatGPT (OpenAI), using the GPT-6 model with Pro reasoning and working autonomously from prompts by Dev D. Goyal, found the proof, wrote its Lean 4 formalization, and wrote a first draft of the paper. Dev D. Goyal chose the problems and asked for proofs, formal verification and revisions, but did not supply proof ideas. Claude Opus 5.5, used with Claude Code, then checked the work independently: it rebuilt the formal proofs from source and confirmed that they rely only on Lean's three standard axioms, recomputed the sequence from its definition compared it with the published terms and tested every claim numerically, and read every step of the proofs. It also wrote the present version of this paper. Dev D. Goyal has reviewed the proofs and is responsible for them; this review is not an independent expert validation. The paper has not been peer reviewed. The section "About this paper" at the end gives the details.

1 Introduction

1.1 The sequence

The *On-Line Encyclopedia of Integer Sequences* (OEIS) [5] is a large public database of integer sequences. Many entries also record *conjectures*: patterns that the contributor observed in the data but did not prove.

This paper is about entry A195194 [1], contributed by Paul D. Hanna in 2011. The entry reads, in part:

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G.f.:  A(x) = INV(x - x*INV(x - 2*x*INV(x - 3*x*INV(x - 4*x*INV(x -
...))))), where INV(F(x)) = series reversion of F(x).

Conjecture:  a(n) is odd iff n is a power of 2.  - Paul D. Hanna, Apr 07
2026
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— OEIS A195194, Paul D. Hanna (name, September 2011; conjecture, April 2026)

Here "G.f." stands for *generating function*: the terms $a(1), a(2), a(3), \ldots$ of the sequence are the coefficients of the power series $A(x) = a(1)\,x + a(2)\,x^2 + a(3)\,x^3 + \cdots$.

Series reversion is the power-series version of an inverse function. If $F(x) = x + f_2x^2 + f_3x^3 + \cdots$, then $\mathrm{INV}(F)$ is the series $V(x) = x + v_2x^2 + \cdots$ for which substituting V into F gives back x: $F(V(x)) = x$. For example, $\mathrm{INV}(x - x^2)$ is the series $x + x^2 + 2x^3 + 5x^4 + 14x^5 + \cdots$ of the Catalan numbers. Section 2 explains this in detail. The entry builds A from an infinite tower of such reversions: in the notation of its examples, $A = \mathrm{INV}(x - x\,F_1)$, where $F_1 = \mathrm{INV}(x - 2x\,F_2)$, where $F_2 = \mathrm{INV}(x - 3x\,F_3)$, and so on without end. It is not obvious what such an infinite tower means; Section 3 gives it a precise meaning.

The first terms, starting at $n = 1$, are 1, 1, 4, 29, 312, 4528, 83736, 1899663, 51453040, ... They grow rapidly.

1.2 The question

Hanna's conjecture says that $a(n)$ is odd exactly when n is one of $1, 2, 4, 8, 16, \ldots$. The table shows the parity of the first terms (1 means odd, 0 means even). The powers of 2 are printed in bold.

n	**1**	**2**	3	**4**	5	6	7	**8**	9	10	11	12	13	14	15	**16**
$a(n)$ mod 2	**1**	**1**	0	**1**	0	0	0	**1**	0	0	0	0	0	0	0	**1**
n	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	**32**
$a(n)$ mod 2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	**1**
n	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
$a(n)$ mod 2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

1.3 What this paper proves

Main results. *(1) The infinite tower of reversions defines a unique sequence of series, one for each level, and in particular a unique series A; cutting the tower off after d levels gives the correct coefficients up to degree $d+1$; this holds for arbitrary weights in place of $1, 2, 3, \ldots$ (Theorem 3.1). (2) The coefficient $a(n)$ is odd if and only if n is a power of 2 (Theorem 3.2).*

The reason for (2) is that, modulo 2, the weight 2 at the second level of the tower vanishes. This cuts the tower off after its first level, and what remains is the equation $A = x + A^2$, which defines the Catalan numbers. Their behavior modulo 2 is classical.

1.4 What was known

The sequence was added to the OEIS in 2011, and the conjecture in April 2026. The entry gives no proof. Its computer program evaluates a tower of finite height, which Theorem 3.1

shows is correct. The parity of the Catalan numbers is classical; see for example [2]. We are not aware of an earlier proof of the conjecture.

1.5 Motivation

Infinitely nested expressions are a common way of describing power series in the OEIS, and they raise a natural question: what does the expression mean? Our first theorem answers this for every tower of reversions of this shape: each additional level fixes one more coefficient, so the coefficients of the truncated towers eventually stop changing. The parity proof then shows that working modulo a number p can turn an infinite object into a finite one, because the weight p in the tower becomes 0. Section 6 explains what this gives for other primes p.

1.6 How to read this paper

Section 2 explains the notions we need: power series, substitution and reversion, remainders, and the Catalan numbers modulo 2. Section 3 states the results precisely, and Section 4 proves them. Section 5 describes how the results were checked, by computer and by a formal proof in Lean. Section 6 discusses the scope of the result and some open questions.

2 Background

This section collects the definitions and facts used later. Readers who know formal power series and congruences can skip it and return when needed.

2.1 Formal power series

A *formal power series* is an expression

$$U(x) = u_0 + u_1 x + u_2 x^2 + u_3 x^3 + \cdots ,$$

determined by its sequence of coefficients $u_0, u_1, u_2, \ldots$; see [7, 6]. No number is ever substituted for x, so questions of convergence do not arise. We write $[x^n]\,U = u_n$ for the coefficient of x^n in U; two series are equal when all their coefficients are equal.

Series are added coefficient by coefficient and multiplied like polynomials:

$$[x^n]\,(UV) = u_0 v_n + u_1 v_{n-1} + \cdots + u_n v_0.$$

Each coefficient of a product is a finite sum. For example, $(1-x)(1+x+x^2+x^3+\cdots) = 1$, so $1+x+x^2+\cdots$ is the *multiplicative inverse* of $1-x$, written $\frac{1}{1-x}$. More generally, every series D with constant term 1 has a unique multiplicative inverse $\frac{1}{D}$, whose coefficients are determined one at a time by $D \cdot \frac{1}{D} = 1$ using only addition, subtraction and multiplication. Whenever a quotient of power series appears in this paper, its denominator has constant term 1, so the quotient is again a power series.

A *commutative ring* is a set with an addition, a subtraction and a multiplication that satisfy the usual rules of arithmetic; examples are $\mathbb{Z}$, $\mathbb{Q}$ and the integers modulo m. For a commutative ring R we write $R[[x]]$ for the set of power series with coefficients in R; for instance, writing $\mathbb{F}_2$ for the integers modulo 2, $\mathbb{F}_2[[x]]$ consists of the series with coefficients modulo 2. We write $U(0)$ for the constant term u_0 of U, and $U = O(x^d)$ when the coefficients of $1, x, \ldots, x^{d-1}$ in U are all zero. Thus $U - V = O(x^d)$ means that U and V have the same coefficient of x^n for

every $n < d$. A series with constant term 0 and coefficient of x equal to 1, that is, a series of the form $x + u_2x^2 + u_3x^3 + \cdots$, is called *normalized*.

2.2 Substituting one series into another

If $V(x)$ has constant term 0, we can substitute V for x in U:

$$U(V(x)) = u_0 + u_1V(x) + u_2V(x)^2 + u_3V(x)^3 + \cdots .$$

This is called *composition*, also written $U \circ V$. It makes sense because $V(x)^k$ has no terms below x^k, so only the finitely many terms with $k \le n$ affect the coefficient of x^n. For example, if $U = x+x^2$ and $V = x^2$, then $U(V(x)) = x^2+x^4$. Composition is associative: $U \circ (V \circ W) = (U \circ V) \circ W$.

Substituting a series into itself repeatedly gives its *iterates*: $U^{\circ 2}(x) = U(U(x))$, $U^{\circ 3}(x) = U(U(U(x)))$, and so on, with $U^{\circ 0}(x) = x$ and $U^{\circ 1} = U$. They must not be confused with *powers* $U^2 = U \cdot U$, $U^3 = U \cdot U \cdot U$. (The OEIS sometimes writes $A^k(x)$ for the k-th iterate. In this paper the small circle always means an iterate, and a plain exponent always means a power.)

2.3 Undoing a substitution

A normalized series can always be "undone" by another normalized series.

Lemma 2.1 (Compositional inverse)**.** *For every normalized series U there is exactly one normalized series V with $U(V(x)) = x$, and this V also satisfies $V(U(x)) = x$. If U has integer coefficients, so does V.*

We call V the *compositional inverse* of U. The OEIS calls it the *series reversion* of U. For example, the compositional inverse of $x - x^2$ is $C(x) = x + x^2 + 2x^3 + 5x^4 + 14x^5 + \cdots$, whose coefficients are the Catalan numbers. Indeed $C - C^2 = x$, which says exactly that substituting C into $x - x^2$ gives x.

Proof. Write $V = x + v_2x^2 + v_3x^3 + \cdots$ with unknown coefficients. In $U(V) = V + u_2V^2 + u_3V^3 + \cdots$, the coefficient of x^n (for $n \ge 2$) equals v_n plus terms that involve only $v_2, \ldots, v_{n-1}$. Setting it equal to 0 therefore determines v_n from the earlier coefficients, using only additions, subtractions and multiplications. So there is exactly one V with $U \circ V = x$, and it has integer coefficients if U does. Applying the same construction to V gives a normalized W with $V \circ W = x$. Then $U = U \circ (V \circ W) = (U \circ V) \circ W = W$, so $V \circ U = V \circ W = x$ as well. □

2.4 Remainders and congruences

For an integer $m \ge 2$ we write $a \equiv b \pmod m$ ("a is congruent to b modulo m") when $a - b$ is divisible by m, that is, when a and b leave the same remainder after division by m. The remainder of a on division by m is called its *residue* modulo m; it is one of $0, 1, \ldots, m-1$.

Addition and multiplication are compatible with congruences, so the residues $0, 1, \ldots, m-1$, added and multiplied modulo m, form a ring, written $\mathbb{Z}/m\mathbb{Z}$. When $m = p$ is a prime we also write $\mathbb{F}_p$. In $\mathbb{F}_2 = \{0, 1\}$ we have $1 + 1 = 0$, so addition and subtraction coincide.

For power series with integer coefficients, $U \equiv V \pmod m$ means that *every* coefficient of $U - V$ is divisible by m. Equivalently, the two series become equal when every coefficient

is replaced by its residue; we call this *reducing* the series modulo m. The key fact is the following.

Reduction modulo m is compatible with all the operations above. To reduce a sum, a product, a substitution, or an inverse, we may reduce the ingredients first and then compute in $\mathbb{Z}/m\mathbb{Z}$. This holds because every coefficient of the result is obtained from finitely many coefficients of the ingredients by adding, subtracting and multiplying, and these operations are compatible with remainders.

Rings. Some statements below hold over every commutative ring, for example $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{Z}/m\mathbb{Z}$. No division is used in them; this matters because in $\mathbb{Z}/m\mathbb{Z}$ division is not always possible (for instance, 2 has no inverse in $\mathbb{Z}/4\mathbb{Z}$).

2.5 The method

The main proofs follow three steps.

1. *Uniqueness.* Show that the defining equation determines the coefficients one at a time, using only addition, subtraction and multiplication. Then the equation has exactly one solution of the required shape, in every commutative ring, including $\mathbb{Z}/2\mathbb{Z}$.
2. *Reduction.* Reducing the integer solution modulo 2 gives a solution of the reduced equation, because reduction is compatible with all operations.
3. *A candidate solution.* Find an explicit series that satisfies the reduced equation. By uniqueness, it is the reduction of the integer solution, and its coefficients can be read off.

Finding the candidate in step 3 is usually the main difficulty.

2.6 The Frobenius identity

Lemma 2.2 (Frobenius identity)**.** *Let p be a prime. For every power series U with coefficients in $\mathbb{F}_p$, we have $U(x)^p = U(x^p)$.*

Proof. In $\mathbb{F}_p$, the binomial theorem simplifies to $(a+b)^p = a^p + b^p$, because every binomial coefficient $\binom{p}{i}$ with $0 < i < p$ is divisible by p. Also $c^p = c$ for every $c \in \mathbb{F}_p$ (Fermat's little theorem). So $\left(\sum_n u_n x^n\right)^p = \sum_n u_n^p x^{pn} = \sum_n u_n x^{pn} = U(x^p)$. The first equality holds for polynomials by repeated use of $(a+b)^p = a^p + b^p$; it holds for power series because each coefficient of $U(x)^p$ involves only finitely many coefficients of U. □

For $p = 2$ this says that squaring a series modulo 2 doubles every exponent: for instance, $(x + x^3)^2 \equiv x^2 + x^6 \pmod 2$. In particular, the square of any series has, modulo 2, no terms of odd degree.

2.7 The Catalan numbers modulo 2

Lemma 2.3. *In every commutative ring the equation $U = x + U^2$ has exactly one solution with $U(0) = 0$. Over the integers it is the Catalan series $C(x) = x + x^2 + 2x^3 + 5x^4 + 14x^5 + \cdots$, and*

$$C(x) \equiv x + x^2 + x^4 + x^8 + \cdots = \sum_{k \geq 0} x^{2^k} \pmod 2.$$

So $[x^n]\,C$ is odd exactly when n is a power of 2.

Proof. The coefficient of x^n in U^2 involves only $u_1, \dots, u_{n-1}$, so the equation determines u_n one at a time: $u_1 = 1$ and $u_n = \sum_{i=1}^{n-1} u_i u_{n-i}$. Over $\mathbb{F}_2$, the series $S = \sum_k x^{2^k}$ satisfies $x + S^2 = x + \sum_k x^{2^{k+1}} = S$ by Lemma 2.2, so by uniqueness $C \equiv S \pmod 2$. □

The parity of the Catalan numbers is classical; see for example [2]. (Here the Catalan number $\frac{1}{n}\binom{2n-2}{n-1}$ appears as the coefficient of x^n, so "n a power of 2" corresponds to the familiar statement that the k-th Catalan number is odd exactly when $k+1$ is a power of 2.)

3 Results

Following the OEIS, we write $\mathrm{INV}(F)$ for the compositional inverse of a normalized series F (Lemma 2.1).

Towers of reversions. Let R be a commutative ring, and let $w_0, w_1, w_2, \dots$ be elements of R, called the *weights*. A *solution of the tower* is a sequence of normalized series $F_0, F_1, F_2, \dots$ over R such that

$$F_j = \mathrm{INV}\big(x - w_j\, x\, F_{j+1}\big) \qquad \text{for every } j \ge 0. \tag{1}$$

Then $F_0 = \mathrm{INV}(x - w_0 x F_1) = \mathrm{INV}\big(x - w_0 x\, \mathrm{INV}(x - w_1 x F_2)\big) = \cdots$. For OEIS A195194 the weights are $w_j = j+1$, so F_0 is the series in the definition of the entry. The *tower of height* d is obtained by replacing the innermost $\mathrm{INV}(\cdots)$ by x after d levels: we define $F_j^{[d]}$ for all j by

$$F_j^{[0]} = x, \qquad F_j^{[d+1]} = \mathrm{INV}\big(x - w_j\, x\, F_{j+1}^{[d]}\big).$$

For example, $F_0^{[2]} = \mathrm{INV}\big(x - w_0 x\, \mathrm{INV}(x - w_1 x^2)\big)$.

Theorem 3.1 (The tower is well defined)**.** *For every commutative ring R and all weights $w_0, w_1, \dots$ in R, the tower* (1) *has exactly one solution. Moreover, the towers of finite height approach it: $F_j^{[d]} - F_j = O(x^{d+2})$ for all $d, j \ge 0$. That is, $F_j^{[d]}$ and F_j have the same coefficients up to degree $d+1$.*

So the infinite tower in the definition of the entry has exactly one meaning, and the sequence A195194 is $a(n) = [x^n]\,F_0$ for the weights $w_j = j+1$ over the integers.

Theorem 3.2 (Hanna's conjecture)**.** *For the weights $w_j = j+1$ over the integers, the coefficient $a(n) = [x^n]\,F_0$ is odd if and only if n is a power of 2. More precisely,*

$$F_0(x) \equiv x + x^2 + x^4 + x^8 + \cdots = \sum_{k \ge 0} x^{2^k} \pmod 2.$$

4 Proofs

The idea in brief. Each reversion in the tower can be rewritten as an ordinary equation: $F_j = \mathrm{INV}(x - w_j x F_{j+1})$ says that $F_j = x + w_j\, F_j \cdot F_{j+1}(F_j)$. The right-hand side, in degree n, only involves coefficients of degree less than n. So all the series F_j can be built at the same time, one degree at a time; this gives existence and uniqueness. Modulo 2, the weight $w_1 = 2$ is 0, which forces $F_1 = x$; then the equation for F_0 becomes the Catalan equation $F_0 = x + F_0^2$.

Lemma 4.1. *Let U and U' be normalized series that agree below degree r, that is, $U - U' = O(x^r)$. Then their compositional inverses $V = \mathrm{INV}(U)$ and $V' = \mathrm{INV}(U')$ also agree below degree r.*

Proof. Since $U(V) = x = U'(V')$, we have $U(V) - U(V') = U'(V') - U(V') = (U' - U)(V')$. The series $U' - U$ has no terms below degree r, and V' starts with x, so $(U' - U)(V') = O(x^r)$. Now suppose that $V - V'$ were not $O(x^r)$, and let $s < r$ be the lowest degree in which V and V' differ. Then

$$U(V) - U(V') = (V - V') + \sum_{k \geq 2} u_k \left(V^k - V'^k\right).$$

Each $V^k - V'^k = (V - V')(V^{k-1} + V^{k-2}V' + \cdots + V'^{k-1})$ has no terms below degree $s + k - 1 \geq s + 1$. So the right-hand side has a nonzero coefficient in degree s, coming from $V - V'$. This contradicts $U(V) - U(V') = O(x^r)$. □

Proof of Theorem 3.1. Rewriting the tower. Let F_j and F_{j+1} be normalized. The series $x - w_j x F_{j+1}$ is normalized too. By Lemma 2.1, $F_j = \mathrm{INV}(x - w_j x F_{j+1})$ holds if and only if substituting F_j into $x - w_j x F_{j+1}$ gives x, that is, $F_j - w_j\, F_j \cdot F_{j+1}(F_j) = x$, or

$$F_j = x + w_j\, F_j \cdot F_{j+1}(F_j). \tag{2}$$

Existence and uniqueness. Both factors of the product $F_j \cdot F_{j+1}(F_j)$ have no constant term. So the coefficient of x^n in this product involves only coefficients of F_j and F_{j+1} of degree less than n. Hence the whole sequence $F_0, F_1, F_2, \ldots$ can be built at the same time, one degree at a time: every F_j has constant term 0 and coefficient 1 in degree 1; and once all the F_j are known below degree n, equation (2) determines the coefficient of x^n in every F_j. This shows that there is exactly one solution. Only additions and multiplications are used, so this works in every commutative ring.

Towers of finite height. We show $F_j^{[d]} - F_j = O(x^{d+2})$ by induction on d, for all j at once. For $d = 0$, both $F_j^{[0]} = x$ and F_j are normalized, so they agree below degree 2. Suppose $F_{j+1}^{[d]} - F_{j+1} = O(x^{d+2})$. Then the series $x - w_j x F_{j+1}^{[d]}$ and $x - w_j x F_{j+1}$ differ by $w_j x\,(F_{j+1} - F_{j+1}^{[d]}) = O(x^{d+3})$. By Lemma 4.1, their inverses $F_j^{[d+1]}$ and F_j also differ by $O(x^{d+3})$. □

Proof of Theorem 3.2. Let $F_0, F_1, \ldots$ be the solution over the integers with $w_j = j + 1$. Reducing all coefficients modulo 2 gives normalized series over $\mathbb{F}_2$ that satisfy (2) with the weights reduced modulo 2, because reduction is compatible with all the operations involved. By the uniqueness in Theorem 3.1 for the ring $\mathbb{F}_2$, these reductions are *the* solution of the tower over $\mathbb{F}_2$. We now work in $\mathbb{F}_2$.

The weight $w_1 = 2$ is 0 in $\mathbb{F}_2$, so (2) for $j = 1$ says $F_1 = x$. Then (2) for $j = 0$, with $w_0 = 1$, says

$$F_0 = x + F_0 \cdot F_1(F_0) = x + F_0 \cdot F_0 = x + F_0^2.$$

This is the Catalan equation, and by Lemma 2.3 its solution over $\mathbb{F}_2$ is $\sum_{k \geq 0} x^{2^k}$. So $a(n)$ is odd exactly when n is a power of 2. □

Notice that, modulo 2, all the deeper levels $F_2, F_3, \ldots$ of the tower play no role at all.

5 Checking the results

A mathematical proof can contain mistakes, so the results were checked in two independent ways.

Numerical check. We evaluated the tower with exact integer arithmetic, once with height 143 and once with height 153. The two results agree below degree 140, as Theorem 3.1 predicts. This reproduces all 20 terms printed in the OEIS entry, and the parity rule of Theorem 3.2 holds for every n with $1 \le n < 140$. The computation was done with the PARI/GP computer algebra system, using code that was written separately from the formal proofs (file `verification/independent_checks.gp` in the accompanying repository [10]). A computation can only test finitely many cases, so it is a check on the definitions and on the indexing, not a proof.

Formal proof in Lean. Lean [8] is a *proof assistant*: a program in which mathematical statements and their proofs are written in a precise formal language, and every step is checked mechanically down to the basic rules of logic. The theorems of Section 3 are proved in Lean 4 (version 4.35.0) on top of Mathlib [9], Lean's library of formalized mathematics. The statements are collected in the file `lean/Entries/A195194.lean` of the repository, in the namespace `OEIS.A195194`. There, for a sequence of weights `w`, the series F_j of the solution is `tail w j`, and the series $F_j^{[d]}$ of the tower of height d is `approximation w d j`. The proofs are in the module `lean/Families/NestedReversion.lean`. The table lists the results covered by the Lean formalization. The Lean formalization does not cover the additional arguments and observations in Section 6.

Statement in this paper	Lean theorem(s)
Theorem 3.1 for arbitrary weights in any commutative ring	`arbitrary_weights_unique`, `arbitrary_finite_depth`
For $w_j = j + 1$: every F_j is normalized, satisfies both inverse identities, and the solution is unique	`all_tails_normalized`, `all_tails_source`, `nested_definition_unique`
$A = F_0$ is normalized and is the inverse of $x - xF_1$	`normalized_source`, `first_tail_reversion`
The tower of height d is correct up to degree $d + 1$	`finite_nest_coefficients`
Theorem 3.2: $F_0 \equiv x + F_0^2 \pmod 2$, and the parity rule	`binary_equation`, `exact_parity`

The Lean project was rebuilt from source, and the axioms on which each of these theorems depends were listed with Lean's `#print axioms` command. Every one depends only on the three standard axioms of Lean's logic (`propext`, `Classical.choice`, `Quot.sound`); no step is left unproved. A formal proof certifies exactly the statement that Lean checks. Whether that statement says what the OEIS entry means is a separate question, so the Lean definitions were also compared with the definition in the OEIS entry (Section 1.1). The principal theorems are stated once more, using only Mathlib's definitions, in the file `lean/Challenges/Catalan.lean`, and Lean's Comparator tool confirms that the proved theorems have exactly these statements.

6 Discussion

Scope. Theorem 3.2 determines the parity of every term, so it settles the conjecture of the entry completely. It says nothing about the size of $a(n)$, which grows very fast, or about

remainders modulo other numbers.

Other primes. The argument of Theorem 3.2 has a general version. Let p be a prime. Modulo p, the weight $w_{p-1} = p$ vanishes, so $F_{p-1} \equiv x$ and the tower becomes finite:

$$F_0 \equiv \mathrm{INV}\Big(x - x\,\mathrm{INV}\big(x - 2x\,\mathrm{INV}(\cdots \mathrm{INV}(x - (p-1)x^2)\cdots)\big)\Big) \pmod{p}.$$

A power series F over $\mathbb{F}_p$ is called *algebraic* if it satisfies a polynomial equation $Q(x, F(x)) = 0$, where Q is a nonzero polynomial in two variables. The innermost series $\mathrm{INV}(x - (p-1)x^2)$ is algebraic; and if U is algebraic, then so is $x - wxU$ for every nonzero weight w, and so is $\mathrm{INV}(U)$ (if $Q(x, U(x)) = 0$, then substituting $V = \mathrm{INV}(U)$ gives $Q(V(x), x) = 0$). So F_0 is algebraic modulo p. By a theorem of Christol, Kamae, Mendès France and Rauzy [3, 4], the remainders $a(n) \bmod p$ then form a *p-automatic* sequence: they can be computed by a machine with finitely many states that reads the digits of n in base p. For $p = 2$ the automaton only checks whether n is a power of 2. We have not worked out the machines for $p \geq 3$, and we leave this as an open question: *for a prime $p \geq 3$, describe the automaton that computes $a(n) \bmod p$ from the base-p digits of n.* (This paragraph is not part of the Lean formalization.)

About this paper

How the work was done. This paper is one of 28 papers on conjectures from the OEIS that came out of an AI research effort directed by Dev D. Goyal [10]. The roles were as follows.

- *Dev D. Goyal* started the project, chose which problems to pursue, asked for proofs, formal verification and revisions, and decided what to publish. Dev D. Goyal did not supply proof strategies or intermediate arguments.
- *ChatGPT* (OpenAI), using the GPT-6 model with Pro reasoning, worked autonomously from those prompts. It searched the OEIS for suitable conjectures, found the proofs, wrote the Lean 4 formalization, and wrote a first draft of this paper.
- *Claude Opus 5.5*, used with Claude Code, checked the work independently. It rebuilt the Lean project from source and confirmed that the theorems listed in Section 5 rely only on Lean's standard axioms; compared the Lean definitions with the OEIS definition; recomputed the sequence with separately written PARI/GP code, compared it with the terms published in the OEIS, and tested every claim of the paper on many terms; and read every step of the proofs. It then wrote the present version of the paper, expanding the explanations and, in places, writing out an argument in more detail or by a more elementary route. The theorems of Section 3 and the proof strategies are ChatGPT's. The Discussion section, including the short arguments in it, was written by Claude Opus 5.5; its observations were checked by computer and are not part of the Lean formalization.

Dev D. Goyal has reviewed the proofs and is responsible for this paper; this review is not an independent expert validation.

Review status. This paper has not been peer reviewed, and it has not been submitted to a journal or to the OEIS.

Priority. An OEIS entry that labels a statement as a conjecture shows only that its contributor did not know a proof; it does not show that no proof existed. During the research only limited searches for earlier work were made. No claim of priority is made.

Corrections. Please report any error in the issue tracker of the repository [10], where the Lean sources and the numerical checks are also available.

Acknowledgements

The sequence and the conjecture are due to Paul D. Hanna [1].

References

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[10] D. D. Goyal, *Proofs and corrections of OEIS conjectures*, repository, 2026. `https://github.com/dvgyl/oeis-proofs`.